The Electromagnetic Form Factor of the ρ-meson on the Lattice

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1. Motivation

The calculation of hadronic form factors on the lattice provides important information about the internal structure of the corresponding states. However, **calculating the form factor for unstable states**, viz. resonances, is not as straightforward, both from a conceptual point of view and in terms of technical implementation. In this work, we specifically consider **the electromagnetic form factor of the ρ-meson**. It is well known that the matrix element of the current between two eigenstates of the lattice Hamiltonian with energies close to the resonance mass does not possess a well-defined infinite-volume limit, even if the Lellouch-Lüscher factors for the external legs are properly taken into account. The reason for this is that **the triangle diagram**, where the external current is hooked on to one of the constituents of the resonance, does not exhibit a regular dependence on the box size.

In this work, we make use of **the external field method (the Feynman-Hellmann method)**, where the system is placed in an external field, coupling to a spatially periodic current. The structure of the finite-volume energy levels is described by the **the modified Lüscher approach**, which enables one to directly extract the short-range part of the interaction with the external field from the fit to the shifted energy levels. This method has the advantage that the separation of the triangle diagram is automatically taken into account, and the short-range interaction has a regular dependence on L, which is exponentially suppressed for large L. Through this work, we provide **a complete framework for the extraction of the form factor from raw lattice data**, when available. We, in addition, provide **a crude estimate of the short-range contribution to the form factor at the lowest order in ChPT**.

2. The Covariant Formulation of NREFT

We work with the the covariant formulation of the pion non-relativistic effective field theory, whose Lagrangian given by

$$\mathcal{L} = \overbrace{\sum_{a=+,0} \phi_a^\dagger 2W(i(v\cdot\partial) - W)\phi_a}^{\text{Pion Lagrangian}} - \overbrace{\Psi_\mu^\dagger \Psi^\mu}^{\text{Rho Lagrangian}}$$

$W = \sqrt{M^2 + \partial^2 - (v\cdot\partial)^2}$ quantisation axis

$$\Psi^\mu = \phi_+ \overset{\leftrightarrow}{W}{}^\mu f(\overset{\leftrightarrow}{W}{}^2)\phi_0,$$
$$W^\mu = v^\mu W + i(\partial^\mu - v^\mu(v\cdot\partial)),$$
$$\overset{\leftrightarrow}{W}{}^\mu = \frac{1}{2}(\vec{W}{}^\mu - \overset{\leftarrow}{W}{}^\mu)$$
$$f(\overset{\leftrightarrow}{W}{}^2) = f_0 + f_1 \overset{\leftrightarrow}{W}{}^2 + \dots$$

Note that the ρ Lagrangian is constructed using a dipion field. The quantity f is related to experimental data through

$$q\cot\delta = -\frac{24\pi\sqrt{P^2}}{q^2 f^2(q^2)}$$

2.1 Including the Electromagnetic Interaction

Gauging the Lagrangian is no longer straightforward due to the presence of non-linear operators. It can be seen that the gauging the quadratic operator leads to

$$W^2 \to W^2 - ie(\partial\cdot A + A\cdot\partial) + ie((v\cdot\partial)(v\cdot A) + (v\cdot A)(v\cdot\partial)) + \mathcal{O}(e^2)$$
$$\equiv W^2 + B + \mathcal{O}(e^2)$$

Then the non-linear operator can be gauged as

$$W \to W + D, \quad D = \int_0^\infty dt\, e^{-tW} B e^{tW}$$

However, the contribution of the 4-pion-photon vertex to the EM form factor of the ρ-meson from this procedure is not apparent. We therefore explicitly include the operators that lead to the relevant form factor structures:

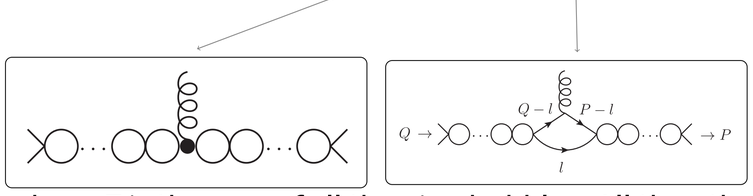
$$\mathcal{L}_{\text{c.t.}} = \sum_{i=1}^{4} g_i e O_i,$$
$$O_1 = \Psi_\mu^\dagger \{\partial^\beta F_{\alpha\beta}, W_\alpha\} g^{\mu\nu}\Psi_\nu,$$
$$O_2 = \Psi_\mu^\dagger i F^{\mu\nu}\Psi_\nu,$$
$$O_3 = \Psi_\mu^\dagger [W^\mu, [W^\nu, \{\partial^\beta F_{\alpha\beta}, W_\alpha\}]]\Psi_\nu.$$
$$O_4 = \Psi_\mu^\dagger [W^\mu, [W^\nu, \{B^\alpha, W_\alpha\}]]\Psi^\nu,$$

where $F^{\alpha\beta} = \partial^\alpha A^\beta - \partial^\beta A^\alpha$ and $B^\alpha = \partial_\beta F^{\alpha\beta}$. These operators are required to construct the Lagrangian in a covariant way.

2.2 Invariant Form Factors

We can now calculate the relevant form factors in the NREFT, and relate them to standard ChPT form factors:

$$\begin{aligned} G_1(k^2) &= -eZg_1k^2 \quad +eZ\langle F_1\rangle \\ G_2(k^2) &= -eZg_2 \quad +eZ\langle F_2\rangle \\ \frac{1}{2s_R}G_3(k^2) &= eZg_3 \quad +eZ\langle F_3\rangle \end{aligned}$$



where Z is the sum of all the pion bubbles. All the other quantities above are also well-defined in the NREFT.

3. The Modified Lüscher Approach

The two-pion is system is placed in a periodic electromagnetic field on a cubic lattice of size L. The two-point function is appropriately modified and it defines the energy spectrum of the two-pion system in a background field. The quantisation condition can also be derived:

$$\det(\Pi_{\mu\nu}^{-1} - D_{\mu\nu}^0) = 0$$

"self-energy" the two-point function of the interpolating field

which determines the position of the poles. The interpolating field is related to the dipion field. Now projecting this on to the relevant irreps and specific background field configurations, leads to relations involving g_1, g_2 and g_3, which can be fit to lattice data. This lets us extract the electromagnetic form factors.

4. Conclusions

The formalism developed here establishes a **consistent way to extract the electromagnetic structure of unstable resonances directly from finite-volume spectra**. In the particular case of the ρ-meson, by combining **the external-field approach** with **the modified Lüscher equation**, we isolate the short-range contribution to the ρ-meson form factors and remove the pathological finite-volume effects originating from the triangle diagram. Additionally, this framework also gives an **estimate of the size of the couplings starting from ChPT**, which is useful before running simulations. Therefore, this framework lays the groundwork for future lattice studies where the extraction of resonance form factors can be performed without resorting to model assumptions or ad hoc subtractions, enabling a clean comparison with phenomenology once lattice data become available.

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